# Canonical Correlation Analysis in Tensor Representation

## Abstract

Canonical Correlation Analysis (CCA) is a multivariate statistical method used to identify relationships between two sets of variables. While traditionally applied to matrix data, modern applications often involve high-dimensional structured data, which can be more naturally represented as tensors. This paper presents a mathematical formulation of CCA in the context of tensor representation, offering a framework for exploring correlations between datasets structured as tensors.

## 1 Introduction

CCA was first introduced by Hotelling in 1936 and has since been widely used in various fields such as signal processing, neuroscience, and machine learning. The goal of CCA is to find linear combinations of variables from two datasets, $\mathbf{X} \in \mathbb{R}^{n\times p}$ and $\mathbf{Y} \in \mathbb{R}^{n\times q}$, such that the correlation between the resulting projections is maximized. Given the growing use of tensor data (multi-dimensional arrays), CCA has been extended to handle tensors in order to exploit the structural information embedded in these multi-way data.

## 2 Notation and Preliminaries

Let us review basic operations on matrices and tensors invoked throughout this paper. Tensors can be considered generalizations of scalars, vectors, and matrices. Let $\mathbf{X}$ represent a $D$-dimensional tensor in $\mathbb{R}^{p_1\times\cdots\times p_D}$. The tensor $\mathbf{X}$ has order $D$, its number of dimensions or modes. For example, vectors are tensors of order one and have one mode. Matrices are tensors of order two and have two modes. We denote an element of $\mathbf{X}$ by $x_{l_1 l_2}, \ldots, l_D$, where $l_i \in [p_i]$ and $i \in [D]$. Fibres are the generalization of matrix rows and columns to higher order tensors. A fibre is defined by fixing the index of every dimension except one dimension. Mode $i$ fibres are $p_i$-dimensional vectors extracted from $X$ by fixing all the indices $(t_1, \ldots, l_{i-1}, l_{i+1}, \ldots, l_D)$ except the $i$ th one $l_i$. Mode $i$ fibres are $p_i$-dimensional vectors extracted from $\mathbf{X}$ by fixing all the indices

$(l_1, \ldots, l_{i-1}, l_{i+1}, \ldots, l_D)$ except the $i$ th one $l_i$. For example, columns of a matrix are Mode 1 fibres, and rows of a matrix are Mode 2 fibres.

It is often useful to reshape a tensor into a matrix. Reordering a tensor into a matrix is referred to as matricization. The mode $i$ matricization of a tensor $\mathbf{X} \in \mathbb{R}^{p_1 \times \ldots \times p_D}$, denoted $\mathbf{X}_{(i)} \in \mathbb{R}^{p_i \times p_{-i}}$ with $p_{-i} = \prod_{k=1,k\neq i}^{D} p_k$, arranges the mode $i$ fibres as the columns of the matrix $\mathbf{X}_{(i)}$. In a mode $i$ matricization, the tensor element $x_{l_1,\ldots,l_D}$ is mapped to the matrix element of $\mathbf{X}_{(i)}$ with index $(l_i, j)$, where $j = 1 + \sum_{k=1,k\neq i}^{D} (l_k - 1) J_k$ with

$$J_k = \begin{cases} 1 & \text{if } k = 1 \text{ or if } k = 2 \text{ and } i = 1 \\ \prod_{k'=1,k'\neq i}^{k-1} p_{k'} & \text{otherwise.} \end{cases}$$

Reordering a tensor into a vector is referred to as vectorization. First describe vectorization of a matrix before describing vectorization of a general tensor. The vectorization of a matrix $\mathbf{X}$ is denoted by $\text{vec}(\mathbf{X})$ and is the vector obtained by stacking the columns of $\mathbf{X}$ on top of each other. The vectorization of the mode $i$ matricization of a tensor $\mathbf{X}$ in turn is denoted as $\mathrm{x}_{(i)} = \text{vec}\left(\mathbf{X}_{(i)}\right)$. We then define the vectorization of a tensor $\boldsymbol{X}$, denoted by $\text{vec}(\mathbf{X})$, as the vectorization of its Mode 1 matricization, namely, $\text{vec}\left(\mathbf{X}_{(1)}\right)$. When unambiguous from context, we will often denote the vectorization of a tensor $\mathbf{X}$ by its corresponding bold lowercase $\mathbf{x}$.

The inner product of two tensors of compatible dimensions $\mathbf{X}, \tilde{X} \in \mathbb{R}^{p_1 \times \ldots \times p_D}$ is the sum of the product of their entries, namely,

$$\langle \mathbf{X}, \tilde{\mathbf{X}} \rangle = \sum_{t_1=1}^{p_1} \ldots \sum_{t_D=1}^{p_D} \mathrm{x}_{l_1,\ldots,t_D} \tilde{\mathrm{x}}_{l_1\ldots,l_D}$$

The mode i product of a tensor $\mathbf{X} \in \mathbb{R}^{p_1 \times \ldots \times p_D}$ with a matrix $\mathbf{U} \in \mathbb{R}^{J \times p_d}$ is denoted by $\mathbf{X} \times_i \mathbf{U}$ and is the tensor of size $p_1 \times \ldots \times p_{i-1} \times J \times p_{i+1} \times \ldots \times p_D$ with elements

$$(\boldsymbol{X} \times_i \mathbf{U})_{l_1,\ldots,l_{i-1}j,l_{i+1},\ldots,t_D} = \sum_{l_i=1}^{p_i} x_{l_1 l_2,\ldots,l_D} u_{j_l i}$$

# 3 Tensor Canonical Correlation Analysis

Let $\mathbf{X} \in \mathbb{R}^{p_1 \times \ldots \times p_{D_x}}$ and $\mathbf{Y} \in \mathbb{R}^{q_1 \times \ldots \times q_{D_y}}$ be two random tensors of order $\mathrm{D_x}$ and $\mathrm{D_y}$, respectively. We denote the vectorizations of $\mathbf{X}$ and $\mathbf{Y}$ by $\mathbf{x}$ and $\mathbf{y}$, respectively. Denote by $\Sigma_{\mathrm{x}}$ and $\Sigma_{\mathrm{y}}$ the covariances of $\mathbf{x}$ and $\mathbf{y}$, respectively. Denote by $\Sigma_{\mathrm{x,y}}$ the covariance between $\mathbf{x}$ and $\mathbf{y}$. Let $\mathbf{V} \in \mathbb{R}^{\mathrm{p}_1 \times \ldots \times \mathrm{p}_{\mathrm{D_x}}}$ and $\mathbf{W} \in \mathbb{R}^{q_1 \times \ldots \times \mathrm{q}_{D_y}}$ be constant tensors, and let $\rho(\mathbf{V}, \mathbf{W})$ denote the correlation between the two linear combinations $\langle \mathbf{X}, \mathbf{V} \rangle$ and $\langle \mathbf{Y}, \mathbf{W} \rangle$, namely,

$$\rho(\mathbf{V}, \mathbf{W}) = \frac{\text{Cov}(\langle \mathbf{X}, \mathbf{V} \rangle, \langle \mathbf{Y}, \mathbf{W} \rangle)}{\sqrt{\text{Var}(\langle \mathbf{X}, \mathbf{V} \rangle)}\sqrt{\text{Var}(\langle \mathbf{Y}, \mathbf{W} \rangle)}} = \frac{\mathbf{v}^\top \Sigma_{\mathbf{x},\mathbf{y}} \mathbf{w}}{\sqrt{\mathbf{v}^\top \Sigma_{\mathbf{x}} \mathbf{v}}\sqrt{\mathbf{w}^\top \Sigma_{\mathrm{y}} \mathbf{w}}} \tag{1}$$

The pair $(\mathbf{V}, \mathbf{W})$ that maximizes $\rho$ are the canonical tensors, and the optimal $\rho$ is the canonical coefficient.

## 4 Discussion

CCA in tensor form allows us to generalize the correlations to multi-dimensional data, retaining the intrinsic relationships across multiple modes (dimensions). This makes Tensor CCA particularly powerful for analyzing complex data, such as those encountered in neuroimaging, genomics, and social network analysis. By projecting tensors instead of matrices, Tensor CCA can capture richer interactions that are not available in traditional CCA models.

## 5 Conclusion

Canonical Correlation Analysis has been extended to tensor data to handle multi-way structures inherent in modern datasets. Tensor CCA offers a more flexible and powerful approach for exploring correlations between higher-order data representations, and it has wide applications in areas involving complex, structured datasets. Further research could focus on computational efficiency and application to specific domains.

## References


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